GE Healthcare

# Signa® 1.5T General Purpose Flex Coil (GE Catalog P/N M1085GR)

## Service Manual

Direction 2131751
Rev 3

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## REVISION HISTORY

| REV | DATE | AUTHOR | REASON FOR CHANGE |
|---|---|---|---|
| 0 | Aug 17, 1998 | L. Loehrer | Converted Toolbook document to MS Word 7.0. |
| 1 | May 20, 1999 | S.M. Atladottir | Updated procedure references for new GUI. |
| 2 | Oct. 24, 2001 | K. LaBarge | Updated procedure for SPT phantom and loader. |
| 3 | May 28, 2010 | R. David | Removed PIN Diode Replacement section and PIN Diode FRU. |

## DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “damage in shipment” written on all copies of the freight or express bill before delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366 and select Option 6.
- Fill out a report at: http://egems.med.ge.com/edq/home.jsp.
- Contact your local service coordinator for more information on this process.

## LANGUAGE POLICY

| | |
|---|---|
| **ПРЕДУПРЕЖДЕНИЕ** (BG) | ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.<br>АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.<br>• НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.<br>• НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ. |
| 警告 (ZH-CN) | 本维修手册仅提供英文版本。<br>• 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。<br>• 未详细阅读和完全理解本维修手册之前，不得进行维修。<br>• 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。 |
| 警告 (ZN-HK) | 本服務手冊僅提供英文版本。<br>• 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。<br>• 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。<br>• 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他的危險。 |
| 警告 (ZH-TW) | 本維修手冊僅有英文版。<br>• 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。<br>• 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。<br>• 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。 |
| **UPOZORENJE** (HR) | OVAJ SERVISNI PRIRUČNIK DOSTUPAN JE NA ENGLESKOM JEZIKU.<br>• AKO DAVATELJ USLUGE KLIJENTA TREBA NEKI DRUGI JEZIK, KLIJENT JE DUŽAN OSIGURATI PRIJEVOD.<br>• NE POKUŠAVAJTE SERVISIRATI OPREMU AKO NISTE U POTPUNOSTI PROČITALI I RAZUMJELI OVAJ SERVISNI PRIRUČNIK.<br>• ZANEMARITE LI OVO UPOZORENJE, MOŽE DOĆI DO OZLJEDE DAVATELJA USLUGE, OPERATERA ILI PACIJENTA USLIJED STRUJNOG UDARA, MEHANIČKIH ILI DRUGIH RIZIKA. |

**VÝSTRAHA**

(CS)

TENTO PROVOZNÍ NÁVOD EXISTUJE POUZE V ANGLICKÉM JAZYCE.

- PŘÍPADĚ, ŽE EXTERNÍ SLUŽBA ZÁKAZNÍKŮM POTŘEBUJE NÁVOD V JINÉM JAZYCE, JE ZAJIŠTĚNÍ PŘEKLADU DO ODPOVÍDAJÍCÍHO JAZYKA ÚKOLEM ZÁKAZNÍKA.
- NESNAŽTE SE O ÚDRŽBU TOHOTO ZAŘÍZENÍ, ANIŽ BYSTE SI PŘEČETLI TENTO PROVOZNÍ NÁVOD A POCHOPILI JEHO OBSAH.
- PŘÍPADĚ NEDODRŽOVÁNÍ TÉTO VÝSTRAHY MŮŽE DOJÍT K PORANĚNÍ PRACOVNÍKA PRODEJNÍHO SERVISU, OBSLUŽNÉHO PERSONÁLU NEBO PACIENTŮ VLIVEM ELEKTRICKÉHO PROUDU, RESPEKTIVE VLIVEM MECHANICKÝCH ČI JINÝCH RIZIK.

**ADVARSEL**

(DA)

DENNE SERVICEMANUAL FINDES KUN PÅ ENGELSK.

- HVIS EN KUNDES TEKNIKER HAR BRUG FOR ET ANDET SPROG END ENGELSK, ER DET KUNDENS ANSVAR AT SØRGE FOR OVERSÆTTELSE.
- FORSØG IKKE AT SERVICERE UDSTYRET MEDMINDRE DENNE SERVICEMANUAL HAR VÆRET KONSULTERET OG ER FORSTÅET.
- MANGLENDE OVERHOLDELSE AF DENNE ADVARSEL KAN MEDFØRE SKADE PÅ GRUND AF ELEKTRISK, MEKANISK ELLER ANDEN FARE FOR TEKNIKEREN, OPERATØREN ELLER PATIENTEN.

**WAARSCHUWING**

(NL)

DEZE ONDERHOUDSHANDLEIDING IS ENKEL IN HET ENGELS VERKRIJGBAAR.

- ALS HET ONDERHOUDSPERSONEEL EEN ANDERE TAAL VEREIST, DAN IS DE KLANT VERANTWOORDELIJK VOOR DE VERTALING ERVAN.
- PROBEER DE APPARATUUR NIET TE ONDERHOUDEN VOORDAT DEZE ONDERHOUDSHANDLEIDING WERD GERAADPLEEGD EN BEGREPEN IS.
- INDIEN DEZE WAARSCHUWING NIET WORDT OPGEVOLGD, ZOU HET ONDERHOUDSPERSONEEL, DE OPERATOR OF EEN PATIËNT GEWOND KUNNEN RAKEN ALS GEVOLG VAN EEN ELEKTRISCHE SCHOK, MECHANISCHE OF ANDERE GEVAREN.

**ENGLISH**

(EN)

THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.

- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

**HOIATUS**
(ET)

SEE TEENINDUSJUHEND ON SAADAVAL AINULT INGLISE KEELES.

- KUI KLIENDITEENINDUSE OSUTAJA NÕUAB JUHENDIT INGLISE KEELEST ERINEVAS KEELES, VASTUTAB KLIENT TÕLKETEENUSE OSUTAMISE EEST.
- ÄRGE ÜRITAGE SEADMEID TEENINDADA ENNE EELNEVALT KÄESOLEVA TEENINDUSJUHENDIGA TUTVUMIST JA SELLEST ARU SAAMIST.
- KÄESOLEVA HOIATUSE EIRAMINE VÕIB PÕHJUSTADA TEENUSEOSUTAJA, OPERAATORI VÕI PATSIENDI VIGASTAMIST ELEKTRILÖÖGI, MEHAANILISE VÕI MUU OHU TAGAJÄRJEL.

**VAROITUS**
(FI)

TÄMÄ HUOLTO-OHJE ON SAATAVILLA VAIN ENGLANNIKSI.

- JOS ASIAKKAAN HUOLTOHENKILÖSTÖ VAATII MUUTA KUIN ENGLANNINKIELISTÄ MATERIAALIA, TARVITTAVAN KÄÄNNÖKSEN HANKKIMINEN ON ASIAKKAAN VASTUULLA.
- ÄLÄ YRITÄ KORJATA LAITTEISTOA ENNEN KUIN OLET VARMASTI LUKENUT JA YMMÄRTÄNYT TÄMÄN HUOLTO-OHJEEN.
- MIKÄLI TÄTÄ VAROITUSTA EI NOUDATETA, SEURAUKSENA VOI OLLA HUOLTOHENKILÖSTÖN, LAITTEISTON KÄYTTÄJÄN TAI POTILAAN VAHINGOITTUMINEN SÄHKÖISKUN, MEKAANISEN VIAN TAI MUUN VAARATILANTEEN VUOKSI.

**ATTENTION**
(FR)

CE MANUEL DE SERVICE N'EST DISPONIBLE QU'EN ANGLAIS.

- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENIR SUR LES EQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ETE CONSULTE ET COMPRIS
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

**WARNUNG**
(DE)

DIESE SERVICEANLEITUNG EXISTIERT NUR IN ENGLISCHER SPRACHE.

- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT DIESE ANLAGE ZU WARTEN, OHNE DIESE SERVICEANLEITUNG GELESEN UND VERSTANDEN ZU HABEN.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH STROMSCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

**ΠΡΟΕΙΔΟΠΟΙΗΣΗ**
(EL)

ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ ΔΙΑΤΙΘΕΤΑΙ ΣΤΑ ΑΓΓΛΙΚΑ ΜΟΝΟ.

- ΕΑΝ ΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ ΕΝΟΣ ΠΕΛΑΤΗ ΑΠΑΙΤΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕ ΓΛΩΣΣΑ ΕΚΤΟΣ ΤΩΝ ΑΓΓΛΙΚΩΝ, ΑΠΟΤΕΛΕΙ ΕΥΘΥΝΗ ΤΟΥ ΠΕΛΑΤΗ ΝΑ ΠΑΡΕΧΕΙ ΥΠΗΡΕΣΙΕΣ ΜΕΤΑΦΡΑΣΗΣ.
- ΜΗΝ ΕΠΙΧΕΙΡΗΣΕΤΕ ΤΗΝ ΕΚΤΕΛΕΣΗ ΕΡΓΑΣΙΩΝ ΣΕΡΒΙΣ ΣΤΟΝ ΕΞΟΠΛΙΣΜΟ ΕΚΤΟΣ ΕΑΝ ΕΧΕΤΕ ΣΥΜΒΟΥΛΕΥΤΕΙ ΚΑΙ ΕΧΕΤΕ ΚΑΤΑΝΟΗΣΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ.
- ΕΑΝ ΔΕ ΛΑΒΕΤΕ ΥΠΟΨΗ ΤΗΝ ΠΡΟΕΙΔΟΠΟΙΗΣΗ ΑΥΤΗ, ΕΝΔΕΧΕΤΑΙ ΝΑ ΠΡΟΚΛΗΘΕΙ ΤΡΑΥΜΑΤΙΣΜΟΣ ΣΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ, ΣΤΟ ΧΕΙΡΙΣΤΗ Ή ΣΤΟΝ ΑΣΘΕΝΗ ΑΠΟ ΗΛΕΚΤΡΟΠΛΗΞΙΑ, ΜΗΧΑΝΙΚΟΥΣ Ή ΑΛΛΟΥΣ ΚΙΝΔΥΝΟΥΣ.

**FIGYELMEZTETÉS**
(HU)

EZEN KARBANTARTÁSI KÉZIKÖNYV KIZÁRÓLAG ANGOL NYELVEN ÉRHETŐ EL.

- HA A VEVŐ SZOLGÁLTATÓJA ANGOLTÓL ELTÉRŐ NYELVRE TART IGÉNYT, AKKOR A VEVŐ FELELŐSSÉGE A FORDÍTÁS ELKÉSZÍTTETÉSE.
- NE PRÓBÁLJA ELKEZDENI HASZNÁLNI A BERENDEZÉST, AMÍG A KARBANTARTÁSI KÉZIKÖNYVBEN LEÍRTAKAT NEM ÉRTELMEZTÉK.
- EZEN FIGYELMEZTETÉS FIGYELMEN KÍVÜL HAGYÁSA A SZOLGÁLTATÓ, MŰKÖDTETŐ VAGY A BETEG ÁRAMÜTÉS, MECHANIKAI VAGY EGYÉB VESZÉLYHELYZET MIATTI SÉRÜLÉSÉT EREDMÉNYEZHETI.

**AÐVÖRUN**
(IS)

ÞESSI ÞJÓNUSTUHANDBÓK ER EINGÖNGU FÁANLEG Á ENSKU.

- EF AÐ ÞJÓNUSTUVEITANDI VIÐSKIPTAMANNS ÞARFNAST ANNAS TUNGUMÁLS EN ENSKU, ER ÞAÐ SKYLDA VIÐSKIPTAMANNS AÐ SKAFFA TUNGUMÁLAÞJÓNUSTU.
- REYNIÐ EKKI AÐ AFGREIÐA TÆKIÐ NEMA AÐ ÞESSI ÞJÓNUSTUHANDBÓK
  HEFUR VERIÐ SKOÐUÐ OG SKILIN.
- BROT Á SINNA ÞESSARI AÐVÖRUN GETUR LEITT TIL MEIÐSLA Á ÞJÓNUSTUVEITANDA, STJÓRNANDA EÐA SJÚKLINGS FRÁ RAFLOSTI, VÉLRÆNU EÐA ÖÐRUM ÁHÆTTUM.

**AVVERTENZA**
(IT)

IL PRESENTE MANUALE DI MANUTENZIONE E DISPONIBILE SOLTANTO IN INGLESE.

- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE E TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER
  CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO
- IL NON RISPETTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

| | |
|---|---|
| 警告<br>(JA) | このサービスマニュアルには英語版しかありません。<br>• サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。<br>• このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。<br>• この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。 |
| 경고<br>(KO) | 본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.<br>• 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.<br>• 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오.<br>• 이 경고에 유의하지 않으면 전기 쇼크, 기계상의 혹은 다른 위험으로부터 서비스 제공자, 운영자 혹은 환자에게 위해를 가할 수 있습니다. |
| **BRĪDINĀJUMS**<br>(LV) | ŠĪ APKALPES ROKASGRĀMATA IR PIEEJAMA TIKAI ANGĻU VALODĀ.<br>• JA KLIENTA APKALPES SNIEDZĒJAM NEPIECIEŠAMA INFORMĀCIJA CITĀ VALODĀ, NEVIS ANGĻU, KLIENTA PIENĀKUMS IR NODROŠINĀT TULKOŠANU.<br>• NEVEICIET APRĪKOJUMA APKALPI BEZ APKALPES ROKASGRĀMATAS IZLASĪŠANAS UN SAPRAŠANAS.<br>• ŠĪ BRĪDINĀJUMA NEIEVĒROŠANA VAR RADĪT ELEKTRISKĀS STRĀVAS TRIECIENA, MEHĀNISKU VAI CITU RISKU IZRAISĪTU TRAUMU APKALPES SNIEDZĒJAM, OPERATORAM VAI PACIENTAM. |
| **ĮSPĖJIMAS**<br>(LT) | ŠIS EKSPLOATAVIMO VADOVAS YRA PRIEINAMAS TIK ANGLŲ KALBA.<br>• JEI KLIENTO PASLAUGŲ TIEKĖJAS REIKALAUJA VADOVO KITA KALBA – NE ANGLŲ, NUMATYTI VERTIMO PASLAUGAS YRA KLIENTO ATSAKOMYBĖ.<br>• NEMĖGINKITE ATLIKTI ĮRANGOS TECHNINĖS PRIEŽIŪROS, NEBENT ATSIŽVELGĖTE Į ŠĮ EKSPLOATAVIMO VADOVĄ IR JĮ SUPRATOTE.<br>• JEI NEATKREIPSITE DĖMESIO Į ŠĮ PERSPĖJIMĄ, GALIMI SUŽALOJIMAI DĖL ELEKTROS ŠOKO,<br>• MECHANINIŲ AR KITŲ PAVOJŲ PASLAUGŲ TIEKĖJUI, OPERATORIUI AR PACIENTUI. |

**ADVARSEL**
(NO)

DENNE SERVICEHÅNDBOKEN FINNES BARE PÅ ENGELSK.

- HVIS KUNDENS SERVICELEVERANDØR TRENGER ET ANNET SPRÅK, ER DET KUNDENS ANSVAR Å SØRGE FOR OVERSETTELSE.
- IKKE FORSØK Å REPARERE UTSTYRET UTEN AT DENNE SERVICEHÅNDBOKEN ER LEST OG FORSTÅTT.
- MANGLENDE HENSYN TIL DENNE ADVARSELEN KAN FØRE TIL AT SERVICELEVERANDØREN, OPERATØREN ELLER PASIENTEN SKADES PÅ GRUNN AV ELEKTRISK STØT, MEKANISKE ELLER ANDRE FARER.

**OSTRZEŻENIE**
(PL)

NINIEJSZY PODRĘCZNIK SERWISOWY DOSTĘPNY JEST JEDYNIE W JĘZYKU ANGIELSKIM.

- JEŚLI DOSTAWCA USŁUG KLIENTA WYMAGA JĘZYKA INNEGO NIŻ ANGIELSKI, ZAPEWNIENIE USŁUGI TŁUMACZENIA JEST OBOWIĄZKIEM KLIENTA.
- NIE PRÓBOWAĆ SERWISOWAĆ WYPOSAŻENIA BEZ ZAPOZNANIA SIĘ I ZROZUMIENIA NINIEJSZEGO PODRĘCZNIKA SERWISOWEGO.
- NIEZASTOSOWANIE SIĘ DO TEGO OSTRZEŻENIA MOŻE SPOWODOWAĆ URAZY DOSTAWCY USŁUG, OPERATORA LUB PACJENTA W WYNIKU PORAŻENIA ELEKTRYCZNEGO, ZAGROŻENIA MECHANICZNEGO BĄDŹ INNEGO.

**AVISO**
(PT-BR)

ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.

- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO ECOMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.

O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

**ATENÇÃO**
(PT-PT)

ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.

- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO ECOMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA
- O NÃO CUMPRIMENTO DESTE AVISO PODE COLOCAR EM PERIGO A SEGURANÇA DO TÉCNICO, DO OPERADOR OU DO PACIENTE DEVIDO A CHOQUES ELÉCTRICOS, MECÂNICOS OU OUTROS.

**ATENȚIE**
(RO)

ACEST MANUAL DE SERVICE ESTE DISPONIBIL NUMAI ÎN LIMBA ENGLEZĂ.

- DACĂ UN FURNIZOR DE SERVICII PENTRU CLIENȚI NECESITĂ O ALTĂ LIMBĂ DECÂT CEA ENGLEZĂ, ESTE DE DATORIA CLIENTULUI SĂ FURNIZEZE O TRADUCERE.
- NU ÎNCERCAȚI SĂ REPARAȚI ECHIPAMENTUL DECÂT ULTERIOR CONSULTĂRII ŞI ÎNȚELEGERII ACESTUI MANUAL DE SERVICE.
- IGNORAREA ACESTUI AVERTISMENT AR PUTEA DUCE LA RĂNIREA DEPANATORULUI, OPERATORULUI SAU PACIENTULUI ÎN URMA PERICOLELOR DE ELECTROCUTARE, MECANICE SAU DE ALTĂ NATURĂ.

**ОСТОРОЖНО!**
(RU)

ДАННОЕ РУКОВОДСТВО ПО ОБСЛУЖИВАНИЮ ПРЕДЛАГАЕТСЯ ТОЛЬКО НА АНГЛИЙСКОМ ЯЗЫКЕ.

- ЕСЛИ СЕРВИСНОМУ ПЕРСОНАЛУ КЛИЕНТА НЕОБХОДИМО РУКОВОДСТВО НЕ НА АНГЛИЙСКОМ, А НА КАКОМ-ТО ДРУГОМ ЯЗЫКЕ, КЛИЕНТУ СЛЕДУЕТ САМОСТОЯТЕЛЬНО ОБЕСПЕЧИТЬ ПЕРЕВОД.
- ПЕРЕД ОБСЛУЖИВАНИЕМ ОБОРУДОВАНИЯ ОБЯЗАТЕЛЬНО ОБРАТИТЕСЬ К ДАННОМУ РУКОВОДСТВУ И ПОЙМИТЕ ИЗЛОЖЕННЫЕ В НЕМ СВЕДЕНИЯ.
- НЕСОБЛЮДЕНИЕ ТРЕБОВАНИЙ ДАННОГО ПРЕДУПРЕЖДЕНИЯ МОЖЕТ ПРИВЕСТИ К ТОМУ, ЧТО СПЕЦИАЛИСТ ПО ОБСЛУЖИВАНИЮ, ОПЕРАТОР ИЛИ ПАЦИЕНТ ПОЛУЧАТ УДАР ЭЛЕКТРИЧЕСКИМ ТОКОМ, МЕХАНИЧЕСКУЮ ТРАВМУ ИЛИ ДРУГОЕ ПОВРЕЖДЕНИЕ.

**UPOZORENJE**
(SR)

OVO SERVISNO UPUTSTVO JE DOSTUPNO SAMO NA ENGLESKOM JEZIKU.

- AKO KLIJENTOV SERVISER ZAHTEVA NEKI DRUGI JEZIK, KLIJENT JE DUŽAN DA OBEZBEDI PREVODILAČKE USLUGE.
- NE POKUŠAVAJTE DA OPRAVITE UREĐAJ AKO NISTE PROČITALI I RAZUMELI OVO SERVISNO UPUTSTVO.
- ZANEMARIVANJE OVOG UPOZORENJA MOŽE DOVESTI DO POVREĐIVANJA SERVISERA, RUKOVAOCA ILI PACIJENTA USLED STRUJNOG UDARA ILI MEHANIČKIH I DRUGIH OPASNOSTI.

**UPOZORNENIE**
(SK)

TENTO NÁVOD NA OBSLUHU JE K DISPOZÍCII LEN V ANGLIČTINE.

- AK ZÁKAZNÍKOV POSKYTOVATEĽ SLUŽIEB VYŽADUJE INÝ JAZYK AKO ANGLIČTINU, POSKYTNUTIE PREKLADATEĽSKÝCH SLUŽIEB JE ZODPOVEDNOSŤOU ZÁKAZNÍKA.
- NEPOKÚŠAJTE SA O OBSLUHU ZARIADENIA SKÔR, AKO SI NEPREČÍTATE NÁVOD NA OBLUHU A NEPOROZUMIETE MU.
- ZANEDBANIE TOHTO UPOZORNENIA MÔŽE VYÚSTIŤ DO ZRANENIA POSKYTOVATEĽA SLUŽIEB, OBSLUHUJÚCEJ OSOBY ALEBO PACIENTA ELEKTRICKÝM PRÚDOM, DO MECHANICKÉHO ALEBO INÉHO NEBEZPEČENSTVA.

| | |
|---|---|
| **ATENCION**<br>(ES) | ESTE MANUAL DE SERVICIO SOLO EXISTE EN INGLES.<br>• SI ALGUN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLES, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCION<br>• NO SE DEBERA DAR SERVICIO TECNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO<br>• LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA. |
| **VARNING**<br>(SV) | DEN HÄR SERVICEHANDBOKEN FINNS BARA TILLGÄNGLIG PÅ ENGELSKA.<br>• OM EN KUNDS SERVICETEKNIKER HAR BEHOV AV ETT ANNAT SPRÅK ÄN ENGELSKA ANSVARAR KUNDEN FÖR ATT TILLHANDAHÅLLA ÖVERSÄTTNINGSTJÄNSTER.<br>• FÖRSÖK INTE UTFÖRA SERVICE PÅ UTRUSTNINGEN OM DU INTE HAR LÄST OCH FÖRSTÅR DEN HÄR SERVICEHANDBOKEN.<br>• OM DU INTE TAR HÄNSYN TILL DEN HÄR VARNINGEN KAN DET RESULTERA I SKADOR PÅ SERVICETEKNIKERN, OPERATÖREN ELLER PATIENTEN TILL FÖLJD AV ELEKTRISKA STÖTAR, MEKANISKA FAROR ELLER ANDRA FAROR. |
| **OPOZORILO**<br>(SL) | TA SERVISNI PRIROČNIK JE NA VOLJO SAMO V ANGLEŠKEM JEZIKU.<br>• ČE PONUDNIK STORITVE STRANKE POTREBUJE PRIROČNIK V DRUGEM JEZIKU, MORA STRANKA ZAGOTOVITI PREVOD.<br>• NE POSKUŠAJTE SERVISIRATI OPREME, ČE TEGA PRIROČNIKA NISTE V CELOTI PREBRALI IN RAZUMELI.<br>• ČE TEGA OPOZORILA NE UPOŠTEVATE, SE LAHKO ZARADI ELEKTRIČNEGA UDARA, MEHANSKIH ALI DRUGIH NEVARNOSTI POŠKODUJE PONUDNIK STORITEV, OPERATER ALI BOLNIK. |
| **DİKKAT**<br>(TR) | BU SERVİS KILAVUZUNUN SADECE İNGİLİZCESİ MEVCUTTUR.<br>• EĞER MÜŞTERİ TEKNİSYENİ BU KILAVUZU İNGİLİZCE DIŞINDA BİR BAŞKA LİSANDAN TALEP EDERSE, BUNU TERCÜME ETTİRMEK MÜŞTERİYE DÜŞER.<br>• SERVİS KILAVUZUNU OKUYUP ANLAMADAN EKİPMANLARA MÜDAHALE ETMEYİNİZ.<br>• BU UYARIYA UYULMAMASI, ELEKTRİK, MEKANİK VEYA DİĞER TEHLİKELERDEN DOLAYI TEKNİSYEN, OPERATÖR VEYA HASTANIN YARALANMASINA YOL AÇABİLİR. |

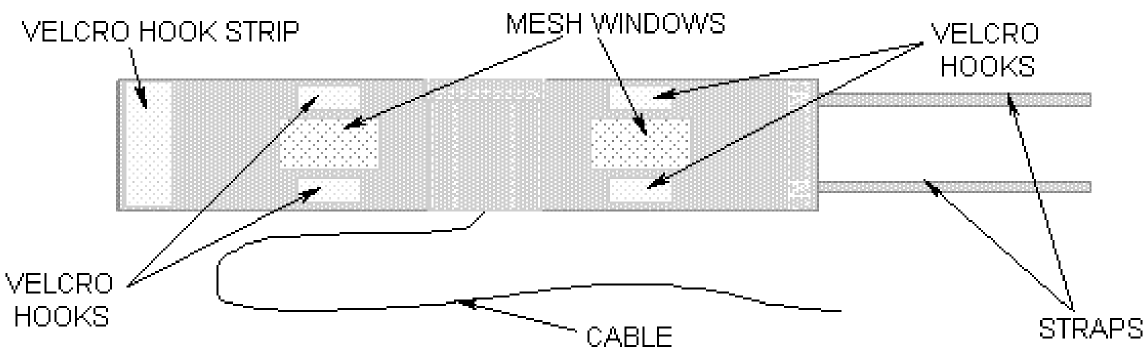
# SECTION 1 - INTRODUCTION

## 1-1 GP Flex Coil Operation

The Signa General Purpose (GP) Flex Coil design is a linear, receive-only flexible coil. The GP Flex Coil consists of two 13 cm x 17 cm (5 in .x 6.5 in.) loops that are serially connected to create a co-rotating "saddle coil" pair. When aligned properly, this "saddle" design provides exceptional uniformity across the sensitive volume of the coil and a minimum of bright spots or high-intensity areas.

If the loops are positioned in a more open fashion (i.e., they move farther away from each other, toward a planar, or flat configuration), the image uniformity decreases. For close positioning do not overlap the coil ends. Doing so will not damage the coil, but the performance will suffer. For optimal positioning, the circular shape should be maintained with the two mesh windows 10 cm (4 in.) apart and the coil ends 5 cm (2 in.) apart. Some deviation from the optimal position will not significantly degrade the uniformity. Therefore, this coil can be used in anatomical areas such as the hip, brachial plexus, larger knees and ankle.

The coil consists of one major assembly shown in Illustration 1-1. The two loops are in a flexible material, covered with fabric. This allows the GP Flex Coil to bend in one direction, making it easy to wrap around the anatomy of interest.



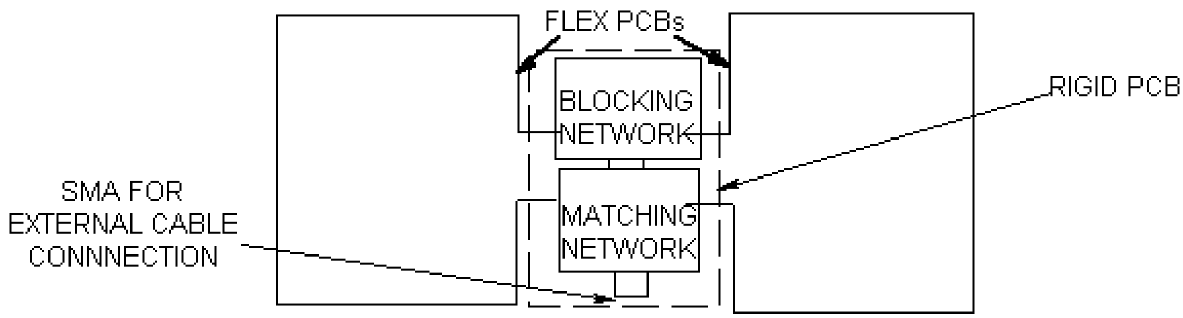

**GP FLEX COIL**
ILLUSTRATION 1-1

The unit is 53 cm (21 in.) long (not including straps) by 21 cm (8 in.) wide. It has two mesh "windows" that indicate the open center of the two coils and aid in positioning.

Two 76 cm (30 in.) long straps are attached to one end of the coil fabric. The undersides of these straps are covered with Velcro that attaches to a Velcro hook strip at the other end of the coil and to four Velcro hooks located above and below the mesh windows.

A 183 cm (72 in.) long RF Cable exits the material on one of the long sides and has a BNC connection that attaches to the Surface Coil quick-disconnect box.

The Functional Block Diagram of the antenna electronics is shown in Illustration 1-2. The two flex PCB wings on either side of the rigid PCB form the loops. The rigid PCB has a matching network that matches the loaded antenna to 50 ohms and an active blocking network. The blocking network consists of a single PIN diode and a tuned LC network. The diode is forward biased whenever the body coil transmits. The external cable is connected to the antenna via a right angle SMA jack.




**GP FLEX COIL FUNCTIONAL BLOCK DIAGRAM**
ILLUSTRATION 1-2

## 1-2 Compatibility

The GP Flex Coil is compatible with the following hardware configurations:

- Signa® 1.5T System
- Signa® Advantage™ 1.5T System
- Signa® Horizon™ 1.5T System

## 1-3 Related Documents

- Direction 15300, Signa® Advantage™ System
- Direction 15304, Signa® Advantage™ 1.5T, 1.0T and 0.5T System
- Direction 15400, Signa® Advantage™ 1.5T, 1.0T and 0.5T System
- Direction 15404, Signa® Advantage™ 1.5T, 1.0T and 0.5T Renewal Parts

## 1-4 Organization of This Document

This manual is divided into five sections:

1- Introduction - Describes how the GP Flex Coil operates and when and where it can be used.

2- Setup and Calibration - Describes installation procedures.

3- Functional Checks - Describes the normal power-up sequence.

4- Replacement/ Maintenance - Describes field maintenance procedures.

5- Renewal Parts - Lists field replaceable parts.

### 1-5 Environmental Requirements

Operate and store the GP Flex Coil in the Scanner Room.

Dimensions: 53 cm x 21 cm x 4 cm (21 in. x 8 in. x 1.5 in.)

## SECTION 2 - SETUP AND CALIBRATION

### 2-1 Checking Shipping List

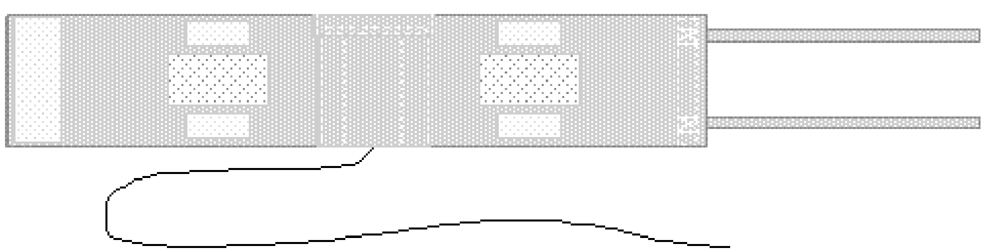
Table 2-1 lists the M1085GR Signa GP Flex Coil parts. Check that all parts were shipped.

TABLE 2-1
**SHIPPING LIST**

| QUANTITY | ITEM | PART NUMBER |
|---|---|---|
| 1 | 1.5T GP Flex Coil | 2128554 |
| 1 | Service Direction | 2131751 |
| 1 | Operator Manual | 2133202 |

### 2-2 Installing GP Flex Coil

1. At the Boot Terminal, install a new soft key. This key will be used by the Operator to select GP Flex Coil imaging. Name the key: **GPFLEX**. (See Illustration 2-1.)



**GP FLEX COIL**
ILLUSTRATION 2-1

2. Refer to the Signa and Signa Advantage System manuals for information on installing soft keys. (Use Coil/Phased Array default values in Table 2-2 and 2-3.)

TABLE 2-2
**COIL VALUES**

| SYSTEM | COIL NAME | COIL TYPE | EXTREMITY COIL | CABLE LOSS | COIL LOSS | RECON SCALE FACTOR | XMIT COIL | MULTI-COIL |
|---|---|---|---|---|---|---|---|---|
| 1.5T, 55cm<br>Low Pass | GPFLEX | SURFACE | NO 1.05 | 1.05 | 0.313 | Head Coil Recon<br>Scale Factor<br>× 0.75 | QUAD | NO |
| 1.5T, 60cm<br>Low Pass | GPFLEX | SURFACE | NO 1.05 | 1.05 | 1.720 | Head Coil Recon<br>Scale Factor<br>× 0.75 | QUAD | NO |

TABLE 2-3
**PHASE ARRAY VALUES**

| NUMBER OF RECEIVERS | START RECEIVER | STOP RECEIVER | PORT ENABLE MASK | ERROR ENABLE MASK |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |

## 2-3 Functional Checks

1. Perform a Body Coil SNR verification. Refer to *Section 3-1, Body Coil SNR Verification*.

2. Perform a GP Flex Coil SNR verification. Refer to *Section 3-2, GP Flex Coil SNR Verification*.

## 2-4 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks outlined below to ensure that the coil is operating properly.

1. Check the external cable for cracks or breaks once a week. Refer to *Section 4-5, Checking the External Cables*.

2. Perform a coil SNR verification. Refer to *Section 3-2, GP Flex Coil SNR Verification*.

3. Record the date and SNR value calculated in *Section 3-3, SNR Image Analysis* in Columns 1 and 2 of the Data Table (found in *Section 6, Forms*) as instructed.

4. Divide the SNR value, obtained in the periodic QA check, by the original SNR value and record in Column 3 of the Data Table.

5. If this ratio is not greater than 85%, there may be a problem in the coil system. Contact your local GE Service Representative.

# SECTION 3 - FUNCTIONAL CHECKS

## 3-1 Body Coil SNR Verification

**Note:**
An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to *System Level Procedures: Troubleshooting: TLT Procedure.*

**Phantom Required:**

- Body TLT Phantom, 46-265635G6 or Body SPT Phantom, 2298119
- Body TLT Loader, 46-287902G1 or SPT Short Loader, 2125244

**Setup Procedure:**

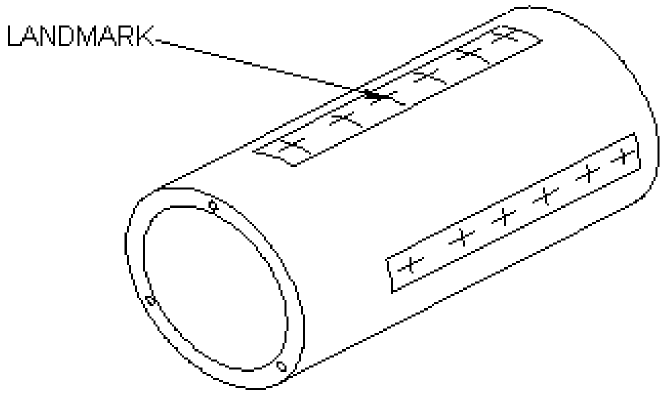
**CAUTION**

**The Quad Head Coil must be completely removed from the cradle before performing any body or surface coil scans. Failure to do this may result in damage to the Quad Head Coil T/R Network.**

1. Remove the Quad Head Coil (if present) from the cradle.

2. Select **[NEW STUDY] (4.x)** or **[New Exam] (5.x)** to allow a new Landmark to be set.

3. Position the Body Phantom in the center of the Body Loader at the center of the cradle. Landmark the center of the phantom and advance to isocenter using the **[ADVANCE TO SCAN]** button. (See Illustration 3-1.)




**BODY PHANTOM/LOADER SETUP**
ILLUSTRATION 3-1

4a. **For 4.x:** Set up scan prescription as shown in Table 3-1.

TABLE 3-1
**BODY COIL SNR - SCAN PROTOCOL (4.X)**

| | | | |
|---|---|---|---|
| ID: | **geservice** | Echo Time (TE): | **[20 msec]** |
| Name: | **body snr** | Rep Time (TR): | **[2000 msec]** |
| Patient Weight: | **300** | Auto CF: | **[Peak]** |
| Patient Entry: | **[Head First]** | Field of View: | **[48 cm]** |
| Patient Position: | **[Supine]** | Scan Thickness: | **[3 mm]** |
| Axial/Sag Landmark: | **[Sternal Notch]** | Scan Location: | **(S/I) 0** |
| Coil Type: | **[Body Coil]** | FOV Center: | **(R/L) R0 (A/P) A0** |
| Scan Plane: | **[Axial]** | Acq. Matrix: | **[256 x 256]** |
| Image Mode: | **[Single Scan]** | Frequency Direction: | **[A/P]** |
| Pulse Sequence: | **[Multiple Echo]** | Imaging Time: | **[1 NEX 8:58]** |
| Imaging Options: | **[None]** | Contrast: | **[No]** |
| or enter PSD filename | | Table Delta: | **0 mm** |
| Number of Echoes: | **[1]** | | |

4b. **For 5.x:** Set up scan prescription as shown in Table 3-2.

TABLE 3-2
**BODY COIL SNR - SCAN PROTOCOL (5.X)**

| | | | |
|---|---|---|---|
| ID: | **geservice** | Rep Time (TR): | **[2000 msec]** |
| Name: | **body snr** | Auto CF: | **[Peak]** |
| Patient Weight: | **300** | Field of View: | **[48 cm]** |
| Patient Entry: | **[Head First]** | Scan Thickness: | **[3 mm]** |
| Patient Position: | **[Supine]** | Interscan Spacing: | **[Other] 0** |
| Axial/Sag Landmark: | **[Sternal Notch]** | Scan Loc (I/S): 0 | **End Loc (S/I): 0** |
| Coil Type: | **[Body Coil]** | No. of Scan Location: | **1** |
| Scan Plane: | **[Axial]** | FOV Center (L/R): | **0 (A/P) 0** |
| Image Mode: | **[2D]** | Acq. Matrix (freq.): | **[256]** |
| Pulse Sequence: | **[Spin Echo]** | Acq. Matrix (phase): | **[256]** |
| Imaging Options: | **[None]** | Frequency Direction: | **[A/P]** |
| or enter PSD filename | | Imaging Time: | **[1 NEX 8:58]** |
| Number of Echoes: | **[1]** | Contrast: | **[No]** |
| Echo Time (TE): | **[20 msec]** | Table Delta: | **0 mm** |

5. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90-degree and 180-degree pulses.

6. Select **[Scan]**. Observe the resulting images to ensure they do not contain artifacts of any sort. Record the Exam number and Series number for SNR Calculations.

7. Select **[Scan]** again. This second image will be used for determination of Body Coil mode SNR.

8. Select **[Cancel]**. Refer to *Section 3-3, SNR Image Analysis* for SNR image analysis.

## 3-2 GP Flex Coil SNR Verification

**Note:**
An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to *System Level Procedures: Troubleshooting: TLT Procedure.*

### Phantom Required:

- NiCl TLT Head Coil Sphere Phantom: P/N 46-265826G6
- Head Coil Loader: P/N 46-287899G1

**WARNING!**

**PHANTOM CONTAINS NICKEL - A SUSPECTED CARCINOGEN. DO NOT INGEST! DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO STATE AND FEDERAL REGULATIONS.**

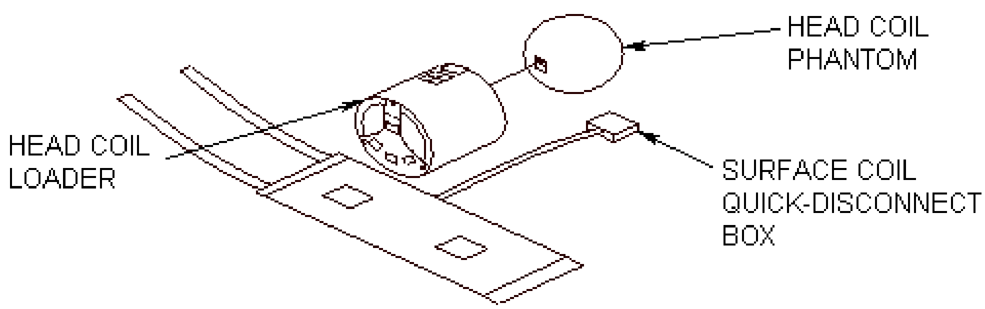
**Setup Procedure:**

1. Select **[NEW STUDY] (4.x)** or **[New Exam] (5.x)** to allow a new Landmark to be set.

**CAUTION**

> **The Quad Head Coil must be completely removed from the cradle before performing any body or surface coil scans. Failure to do this may result in damage to the Quad Head Coil T/R Network.**

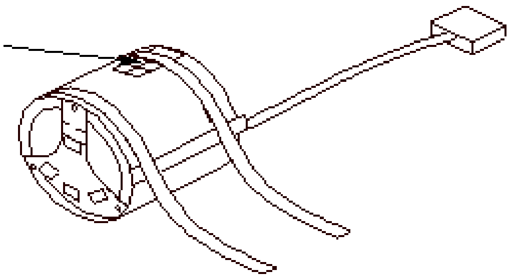
2. Remove the Quad Head Coil (if present) from the cradle.
3. Connect the GP Flex Coil to the surface coil Quick Disconnect box, and plug the box into the carriage assembly.
4. Lay the GP Flex Coil, "Patient Side" up, on the table. Use the alignment lights to center the coil.
5. Place the Head Coil phantom in the loader. (See Illustration 3-2.)




**GP FLEX COIL PHANTOM SETUP**
ILLUSTRATION 3-2

6. Center the loader on the Flex Coil using the alignment lights and the markings on the top of the Head Coil Loader.
7. Secure the coil onto the loader using the coil straps. Use foam wedges on the sides to stabilize the coil and phantom.

8. Landmark using the crosshairs on the phantom (Illustration 3-3), and **[ADVANCE TO SCAN]**.




**GP FLEX COIL PHANTOM LANDMARK**
ILLUSTRATION 3-3

9a. **For 4.x:** Set up scan prescription as shown in Table 3-3.

TABLE 3-3
**GP FLEX COIL SNR - SCAN PROTOCOL (4.X)**

| | | | |
|---|---|---|---|
| ID: | **geservice** | Echo Time (TE): | **[20 msec]** |
| Name: | **gp flex snr** | Rep Time (TR): | **[2000 msec]** |
| Patient Weight: | **300** | Auto CF: | **[Peak]** |
| Patient Entry: | **[Head First]** | Field of View: | **[48 cm]** |
| Patient Position: | **[Supine]** | Scan Thickness: | **[3 mm]** |
| Axial/Sag Landmark: | **[Sternal Notch]** | Scan Location: | **(S/I) 0** |
| Coil Type: | **[Other Coil] [GPFLEX]** | FOV Center: | **(R/L) R0 (A/P) A0** |
| Scan Plane: | **[Axial]** | Acq. Matrix: | **[256 x 256]** |
| Image Mode: | **[Single Scan]** | Frequency Direction: | **[A/P]** |
| Pulse Sequence: | **[Multiple Echo]** | Imaging Time: | **[1 NEX 8:58]** |
| Imaging Options: | **[None]** | Contrast: | **[No]** |
| or enter PSD filename | | Table Delta: | **0 mm** |
| Number of Echoes: | **[1]** | | |

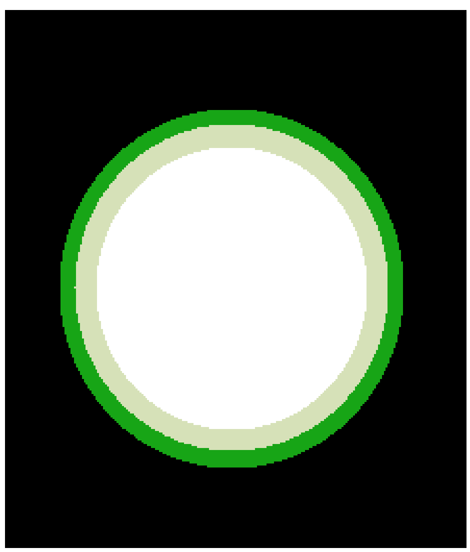
9b. **For 5.x:** Set up scan prescription as shown in Table 3-4.

TABLE 3-4
**GP FLEX COIL SNR - SCAN PROTOCOL (5.X)**

| | | | |
|---|---|---|---|
| ID: | **geservice** | Rep Time (TR): | **[2000 msec]** |
| Name: | **gp flex snr** | Auto CF: | **[Peak]** |
| Patient Weight: | **300** | Field of View: | **[48 cm]** |
| Patient Entry: | **[Head First]** | Scan Thickness: | **[3 mm]** |
| Patient Position: | **[Supine]** | Interscan Spacing: | **[Other] 0** |
| Axial/Sag Landmark: | **[Sternal Notch]** | Scan Loc (I/S): 0 | **End Loc (S/I): 0** |
| Coil Type: | **[Other Coils][GPFLEX]** | No. of Scan Location: | **1** |
| Scan Plane: | **[Axial]** | FOV Center (L/R): | **0 (A/P) 0** |
| Image Mode: | **[2D]** | Acq. Matrix (freq.): | **[256]** |
| Pulse Sequence: | **[Spin Echo]** | Acq. Matrix (phase): | **[256]** |
| Imaging Options: | **[None]** | Frequency Direction: | **[A/P]** |
| or enter PSD filename | | Imaging Time: | **[1 NEX 8:58]** |
| Number of Echoes: | **[1]** | Contrast: | **[No]** |
| Echo Time (TE): | **[20 msec]** | Table Delta: | **0 mm** |

10. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90-degree and 180-degree pulses.

11. Select **[Scan]**, and observe the resulting image of the sphere. (Illustration 3-4 shows a normal image.) Ensure there are **no** artifacts of any sort in the sphere image. Record the Exam number and Series number for SNR calculations.



**GP FLEX COIL IMAGE**
ILLUSTRATION 3-4

12. Select **[Scan]** again. This second image of the sphere will be used to determine the GP Flex Coil mode SNR.

13. Select **[Cancel]**. Refer to *Section 3-3, SNR Image Analysis* for SNR image analysis.

## 3-3 SNR Image Analysis

### 3-3-1 SNR Image Analysis (Release 4.x)

See Section 3-3-2 for Release 5.x.

**Description:**

The CLIPS macro will ask for the first image in the series to analyze. The macro assumes two back-to-back scans were done, and will take the designated image and the next image in the series (i.e., if image 3 is provided, it will take 3 and 4). It will also ask that three points on the border of the phantom image be designated to use in positioning the analysis ROI. If the image is too big, too small, or the three points were selected too close together, the macro will inform you.

**Procedure:**

1. Touch **[UTILITIES]**, then **[Clips]**.

```
      CLIPS
  -------------------------
    1 Run CLIPS
    2 Save files
    3 Restore files
    4 Remove files
    5 Exit
```

`Enter the number of your selection: ..........................` **1 [ENTER]**

```
                       1 = Operator console
                       2 = Remote1
                       3 = Remote2
                       4 = Exit
```

`Which image processor would you like to use (1 2 or 3) ? ......` **1 [ENTER]**

`IP selected is at the operator console.`

`Do you wish to boot the Image Processor? (Y or N) .............` **Y [ENTER]**

**Note:** It is not necessary to reboot the IP each time you enter.

`Welcome to the Command Line Image Processing System (CLIPS)`

`CLIPS > ..............................................` **LIST(STUDY); [ENTER]**

```
                              STUDY LIST
========================================================================
   STUDY #        PATIENT NAME        PATIENT I.D.        DATE
------------------------------------------------------------------------
    00001            BDY_SNR            GESERVICE       9-JULY-89
    00002            HD_SNR             GESERVICE       9-JULY-89
------------------------------------------------------------------------

CURRENT DEFAULTS --- PATIENT ID =                     /  /

                      N - NEXT PAGE
                      P - PREVIOUS PAGE
                      S - SELECT STUDY
                      C - CANCEL
```

choice ?...............................................................**S [ENTER]**
ENTER STUDY NUMBER ...........................................**1 [ENTER]**

**Note:** Enter appropriate number.

CLIPS > ...........................................**LIST(SERIES); [ENTER]**

```
                            SERIES LIST
========================================================================
 SERIES #         # OF IMAGES             SERIES TYPE      PULSE TYPE
------------------------------------------------------------------------
   001              002                       AXIAL              ME
   002              002                       SAG                ME
------------------------------------------------------------------------
CURRENT DEFAULTS --- PATIENT ID = SNR           0000S/00S/001
                             N - NEXT PAGE
                             P - PREVIOUS PAGE
                             S - SELECT SERIES
                             C - CANCEL
```

choice ? ..........................................................**S [ENTER]**
ENTER SERIES NUMBER ..........................................**1 [ENTER]**

**Note:** Enter appropriate number.

CLIPS > ......................................................**DIS; [ENTER]**

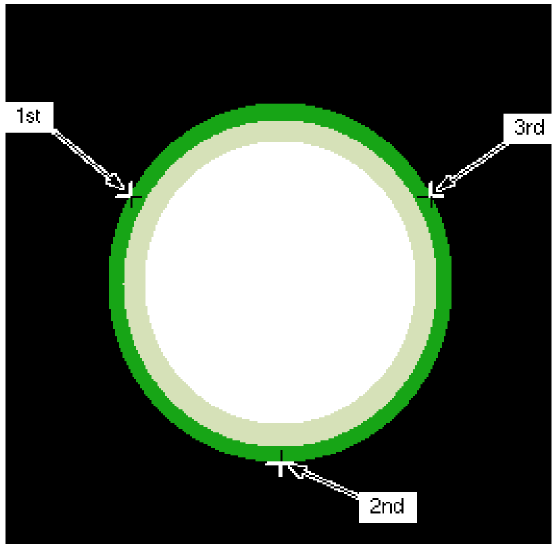
**or: LIST (IMAGE)** to select the desired image.

CLIPS > ........................**EXECUTE(SNR); [ENTER]** or **EXE(SNR); [ENTER]**

```
THIS MACRO CALCULATES THE SNR USING THE GE HEAD OR BODY PHANTOM.
ENTER THE NUMBER OF THE FIRST IMAGE OF THE TWO-IMAGE SET AND !
EXAMPLE INPUT> 1!
Please start entering data.  Type an ! to end input mode.
```

CLIPS INPUT> ...........................................**1! [ENTER]**

Correct positioning of the SNR analysis ROI depends on finding the center of the phantom. You will be prompted to identify three points at the edge of the phantom. Care should be taken to pick three points as far away from each other as possible (i.e., 10, 6 and 2 o'clock) and to avoid flat spots and irregularities on the edge. (See Illustration 3-5.)




**SNR CURSOR POSITIONS**
ILLUSTRATION 3-5

**Note:**
The order of cursor placement must be 10, 6, then 2 o'clock positions. Using a different order may give an error.

`POSITION CURSOR AT FIRST POINT AND ENTER !`

**Note:** Place cursor at 10:00 position on edge of image.

`Please start entering data.  Type an ! to end input mode.`
`CLIPS INPUT> ..........................................` **! [ENTER]**

`POSITION CURSOR AT SECOND POINT AND ENTER !` Place cursor at 6:00 position.

`Please start entering data.  Type an ! to end input mode.`
`CLIPS INPUT> ..........................................` **! [ENTER]**

`POSITION CURSOR AT THIRD POINT AND ENTER !` Place cursor at 2:00 position.

`Please start entering data.  Type an ! to end input mode.`
`CLIPS INPUT> ..........................................` **! [ENTER]**
`PHANTOM POINTS AND RADIUS OK.  BEGIN ANALYSIS.`

**Note:** The program will then perform the analysis and display the final values on the image monitor. Record Signal, Noise, and Signal to Noise in Data Table in Section 6.

```
DO YOU WANT TO SAVE THIS DATA?
ENTER A  1! FOR YES, A  0! FOR NO.
Please start entering data.  Type an ! to end input mode.
CLIPS INPUT>.................................................. 0! [ENTER]

DATA NOT SAVED
CLIPS >  ...........................................LIST(SERIES); [ENTER]
```

**Note:** Select the next series and repeat the above analysis for each image pair. Type **BYE** when completed to exit CLIPS.

2. When all analyses are complete, reset Auto Center Frequency mode by touching **[SCAN MODES]**, **[Default Auto CF]**, then **[EXECUTE]**.

   **Note:**
   It is not necessary to reset the Auto Center Frequency mode if proceeding onto more image quality scans.

3. Record the date and the calculated SNR value in Columns 1 and 2 of the Data Table (found in *Section 6, Forms*) as instructed.

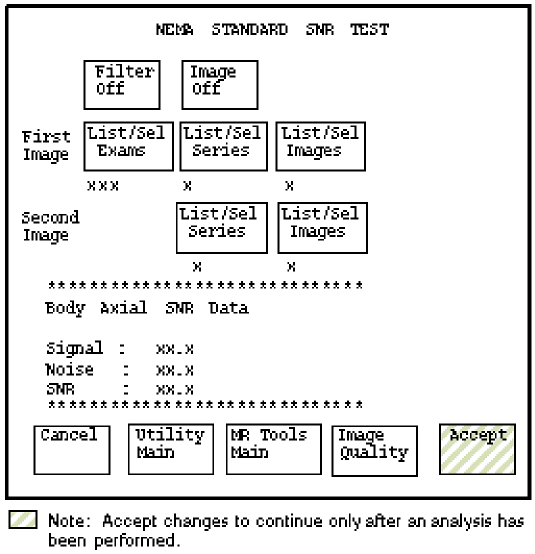
### 3-3-2 SNR Image Analysis (Release 5.x)

**Description:**

The SNR tool retrieves two operator-selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center of the image for positioning the ROI. A difference image is created by subtracting the second image from the first, and the same ROI is used to calculate noise from the subtracted image. The Signal value, Noise value, and Signal-to-Noise ratio are reported. There is an option to save the difference image with the results annotated.

**Procedure:**

1. Touch **[UTILITIES]**, **[MR Tools]**, **[Image Quality]**, then **[SNR Test]**. The SNR Test screen (shown in Illustration 3-6) displays.




**SNR TEST SCREEN**
ILLUSTRATION 3-6

2. Enter image exam, series, and image numbers. If exam, series, or image numbers are not known, select **[List Exams]**, **[List Series]**, or **[List Images]** to display the list to choose from.

   **Note:**
   Image number selection must be back lit (highlighted) to be able to enter information. Use the Switch key on the keyboard to transfer control from the left to right side of Touch Screen.

3. If high pass filtering is desired to be performed on data, touch **[Filter Off]**, which will highlight and change to **[Filter On]**.

4. If the difference image annotated with data is to be created, touch **[Image Off]**, which will highlight and change to **[Image On]**.

5. Touch **[Accept]** to begin analysis. The final values are displayed on the Touch screen. (See Illustration 3-6, which shows the previous screen.)

6. Touch **[Continue]**, select the next exam, and repeat the above analysis for each image pair.

7. Record the date and the calculated SNR value in Columns 1 and 2 of the Data Table (found in *Section 6, Forms*) as instructed.

## 3-4 Checking PIN Diodes with Digital Multimeter (DMM)

**Note:**
There is only one PIN diode in this antenna. This procedure will indicate if the PIN diode is defective.

1. Select the **DIODE TEST** function on the Digital Multimeter (DMM).
2. Connect the “positive“ lead of the DMM to the center pin of the BNC on the external cable, and connect the “negative” lead to the main housing/ground of the BNC.

   **Note:** A reading of 0.400 to 0.600 should be observed on the DMM.
3. If a reading below 0.400 is observed in either direction, the output cable is shorted or the PIN diode is bad.
4. If a reading above 0.600 is observed, the PIN diode is defective.

   **Note:** Due to safety hazards arising from soldering in the field, PIN diodes are no longer FRUs. If the coil has a defective PIN diode, the respective coil component must be replaced.
5. Connect the “positive” lead of the DMM to the main housing/ground of the BNC on the external cable, and connect the “negative” lead to the center pin of the BNC.

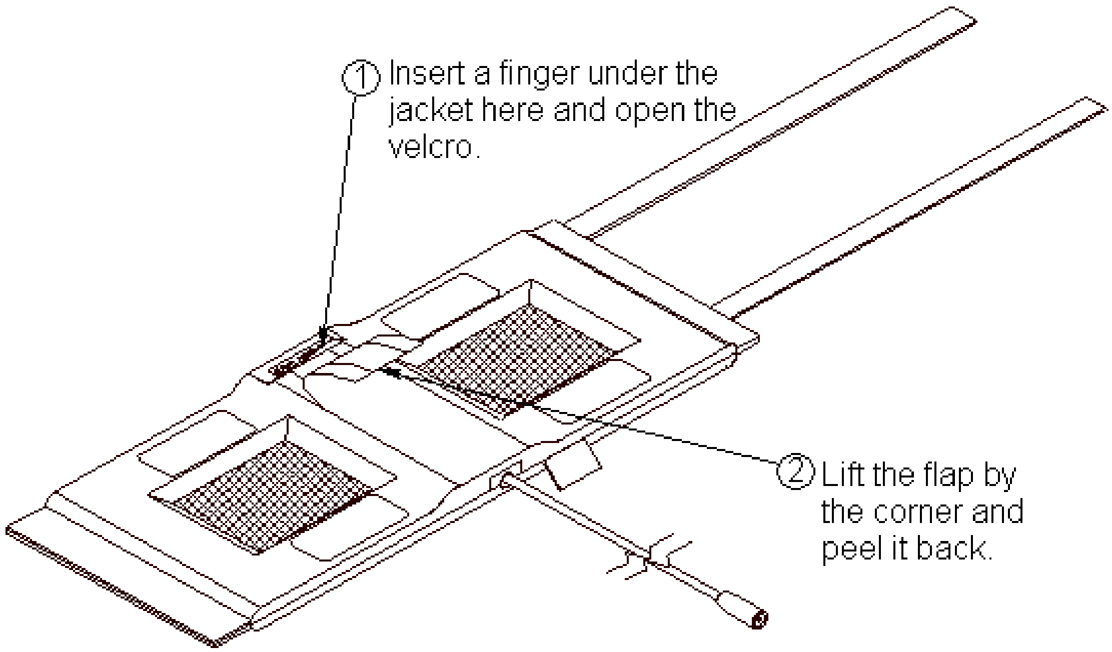
   **Note:** A reading of INFINITY should be observed on the DMM.
6. If a reading of INFINITY is observed in both directions, the output cable is open or the PIN diode is open.

## 3-5 Checking External Cable

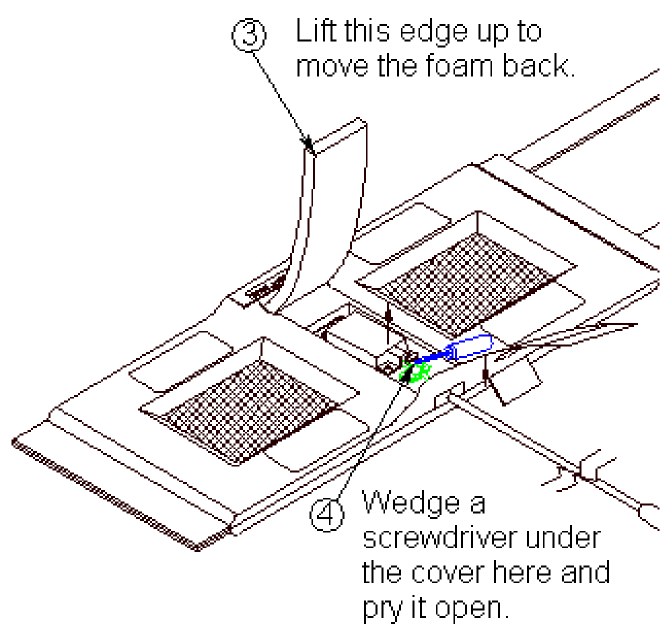
1. Select the **DIODE TEST** function on the DMM.
2. Connect the “positive” lead of the DMM to the center pin of the BNC on the external cable, and connect the “negative” lead to the main housing/ground of the BNC.
3. Flex the external cable, especially near the connectors and the strain relief, and observe that a reading of 0.400 to 0.600 remains on the DMM with no instability or fluctuations.
4. Connect the “positive” lead of the DMM to the main housing/ground of the BNC on the external cable, and connect the “negative” lead to the center pin of the BNC.
5. Flex the external cable, especially near the connectors and the strain relief, and observe that a reading of INFINITY remains on the DMM with no instability or fluctuations.
6. If the cable fails any of the above tests, replace it. Refer to *Section 4-2, Replacing External Cable*.

## SECTION 4 - REPLACEMENT / MAINTENANCE

### 4-1 Disassembly / Assembly of GP Flex Coil

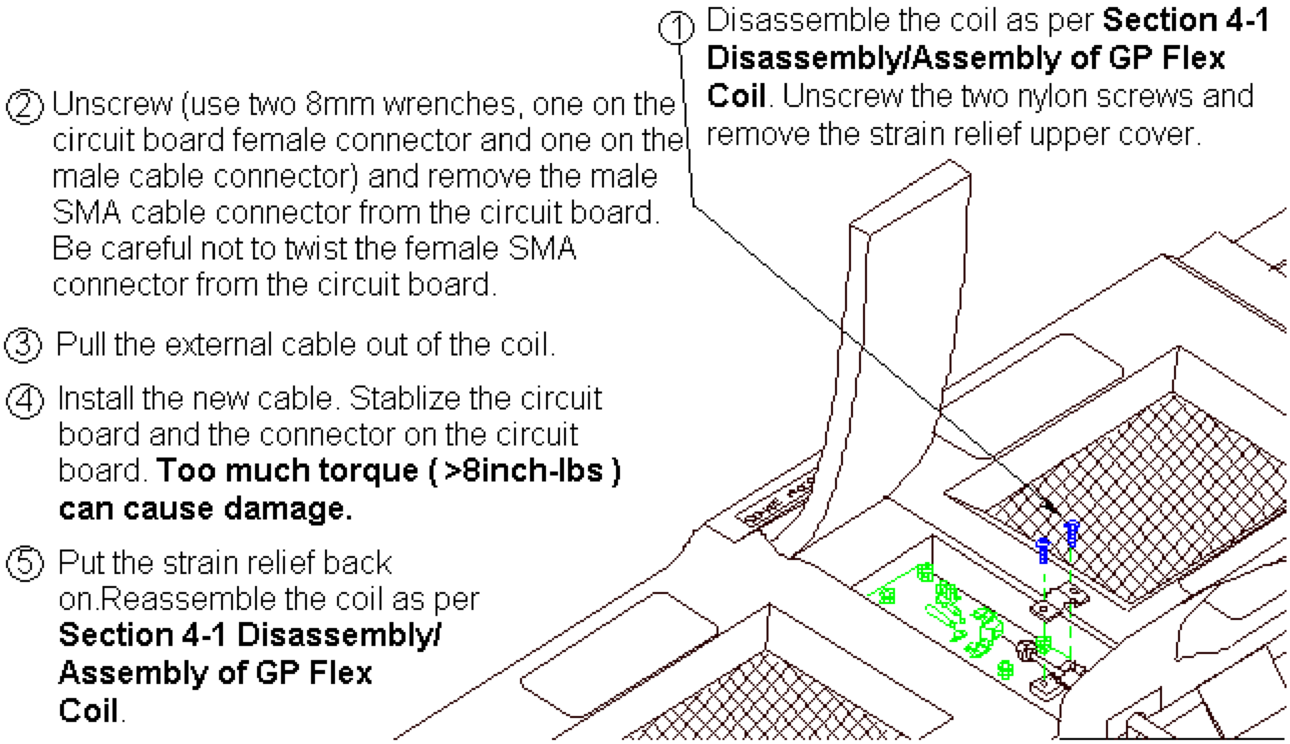



5 To reassemble, place the cover back on the posts and press on it until it snaps into place. Push the foam flap under the jacket. Carefully position the fabric flap in place making sure that the Velcro® on the edges lines up with the Velcro on the jacket. Tuck the end of the flap under the jacket. Press over the Velcro, making sure that no edges are left open.




**REPLACEMENT / MAINTENANCE**
ILLUSTRATION 4-1

## 4-2 Replacing External Cable




**REPLACING EXTERNAL CABLE**
ILLUSTRATION 4-2

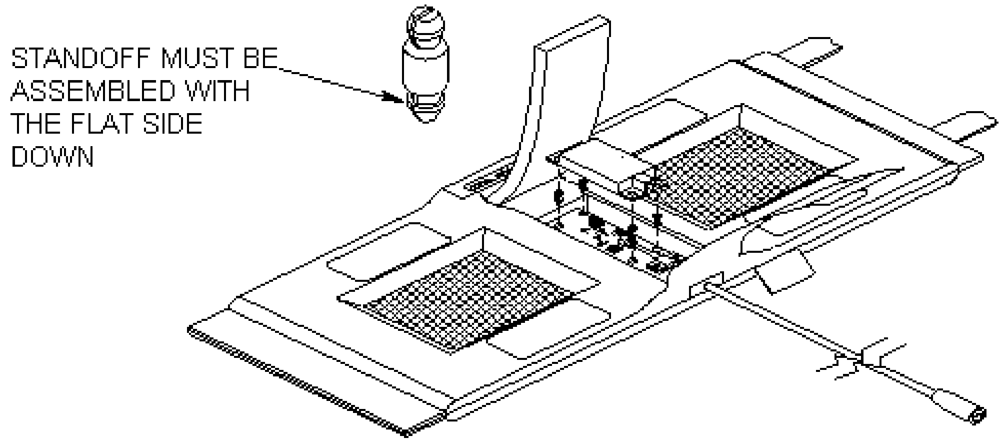
## 4-3 Replacing Mechanical Hardware

1. Refer to *Section 5, Renewal Parts* for the Hardware Kit part number.

2. To replace the cover assembly, refer to *Section 4-1, Disassembly/Assembly of GP Flex Coil*.

3. To replace the upper strain relief block and the two screws refer to *Section 4-2, Replacing External Cable*.

4. If a plastic standoff breaks, it can be pulled out using a needle-nose pliers. Insert a new standoff in its place, making sure that the flat end goes in first. (See Illustration 4-3.)




**STANDOFF REPLACEMENT**
ILLUSTRATION 4-3

## 4-4 Checking External Cable

1. Inspect the external cables for cracks or breaks once each week. Replace the external cable per *Section 4-2, Replacing External Cable* if any damage or wear is found. (See *Section 5, Renewal Parts* for Cable part number.)

## 4-5 Cleaning Coil

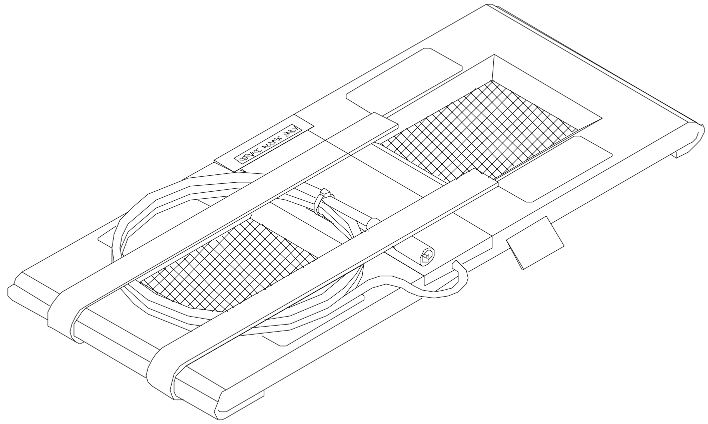
**CAUTION**

**Avoid damaging sensitive electronic parts. Do not spray or pour any solution directly onto the GP Flex Coil or external cable. Do not use alcohol or disinfectant or any chemical cleaner, except mild liquid detergents, to clean the GP Flex Coil. Avoid using bleach solution, as it will degrade the fabric jacket. Never submerge the GP Flex Coil in any liquid.**

Clean the GP Flex Coil and external cable with a mild liquid detergent and water solution. Wet a soft cloth with the solution and proceed to clean.

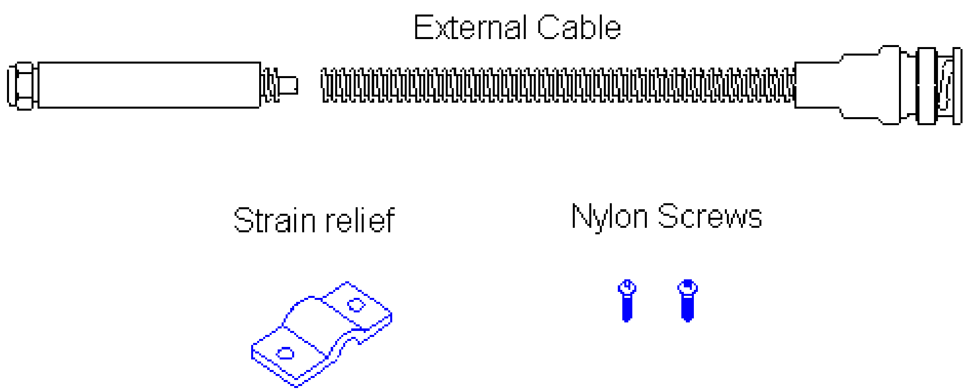
## 5- RENEWAL PARTS

### 1.5T SIGNA GP FLEX COIL WITH CABLE, P/N 2128554



| ITEM | PART NO. | FRU | NAME | QUANTITY | DESCRIPTION |
|---|---|---|---|---|---|
| 1 | 2128554 | 1 | 1.5T GP FLEX COIL | 1 | FOR 1.5T SIGNA GP FLEX COIL |

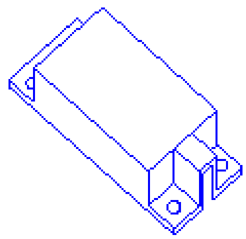
### 1.5T SIGNA GP FLEX COIL CABLE KIT, P/N 2128557



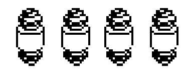

| ITEM | PART NO. | FRU | NAME | QUANTITY | DESCRIPTION |
|---|---|---|---|---|---|
| 1 | 2128557 | 1 | CABLE KIT | 1 | FOR 1.5T SIGNA GP FLEX COIL |

## 1.5T SIGNA GP FLEX COIL HARDWARE KIT, P/N 2128557-2

Cover Assembly



Standoffs



| ITEM | PART NO. | FRU | NAME | QUANTITY | DESCRIPTION |
|---|---|---|---|---|---|
| 1 | 2128557-2 | 2 | HARDWARE KIT | 1 | FOR 1.5T SIGNA GP FLEX COIL |

## SECTION 6 – FORMS

# DATA TABLE

Use the space provided below to record the calculated Signal-to-Noise ratio (SNR) data obtained from *Section 3, Functional Checks*. Record the SNR data obtained during the initial coil installation on the lines indicated below. Record subsequent SNR data in the Data Table as a periodic QA check. If the ratio of any of the coils found is **not** greater than 85%, there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

GP Flex Coil SNR Value: ________________ Date: ______________

### SNR Data QA (Quality Assurance) Check

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| Date | SNR Value | Divide the SNR value from Column 2 by the original SNR. | Is the value in Column 3 > 85%? |
| __________ | __________ | __________ | Y / N |
| __________ | __________ | __________ | Y / N |
| __________ | __________ | __________ | Y / N |
| __________ | __________ | __________ | Y / N |
| __________ | __________ | __________ | Y / N |
| __________ | __________ | __________ | Y / N |
| __________ | __________ | __________ | Y / N |
| __________ | __________ | __________ | Y / N |
| __________ | __________ | __________ | Y / N |

**Note**: Make additional copies of this document as is needed.

# DEFECTIVE COIL RETURN FORM

**Note:**
To allow for proper assessment of defective returned coils, this form must be completely filled out and accompany all returned coils. Include films or prints of any image quality related complaints with a description of scan protocol used.

Date ______________________________________________

Site Name ______________________________________________

Site Address ______________________________________________

Service Engineer ______________________________________________

Coil Serial Number______________________________________________

Date Coil Installed ______________________________________________

Description of Coil Problem ______________________________________________

______________________________________________

______________________________________________

______________________________________________

______________________________________________

______________________________________________

# ELECTRICAL CHECKS

## PIN Diode Test

Meter Reading Forward Bias ______________________

Meter Reading Reverse Bias ______________________

**Note:**
An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to *System Level Procedures: Troubleshooting: TLT Procedure.* Use Table DR-1 and DR-2 to record the TLT results of the defective coil.

TABLE DR-1
**TLT DATA FOR DEFECTIVE SURFACE COIL**

**SITE:** ________________________________ **NAME:** ________________________

**DATE:** ______________________

**TLT FILE NUMBER:** _________________________

**SNR:** NOISE _________________________

SNR MEAN _________________________

SNR SIGNAL _________________________

SNR AREA _________________________

**TR MAP:** 89-91 _________________________ %

85-95 _________________________ %

65-115 _________________________ %

FLIP MEAN _________________________

TABLE DR-2
**TLT DATA FOR DEFECTIVE PHASED ARRAY COIL (NOT APPLICABLE)**

**SITE:** ______________________________ **NAME:** ______________________________

**DATE:** ________________________

**TLT FILE NUMBER:** __________________________

**SNR:** NOISE (0)___________________________

NOISE (1)___________________________

NOISE (2)___________________________

NOISE (3)___________________________

NOISE (C)___________________________

SNR MEAN (0)___________________________

SNR MEAN (1)___________________________

SNR MEAN (2)___________________________

SNR MEAN (3)___________________________

SNR MEAN (C)___________________________

**TR MAP:** 89-91 (C)___________________________ %

(Combined 85-95 (C)___________________________ %

Results) 65-115 (C)___________________________ %

FLIP MEAN (C)___________________________

FLIP SDV (C)___________________________

RCV MEAN (C)___________________________

RCV SDV (C)___________________________

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